KAIZEN MATHS

Lesson starter

Equation of a Circle

Centre and Radius

Name _____

S1

Find the centre and radius of the circle:

$(x + 4)^2 + y^2 = 4$

S2

Find the centre and radius of the circle:

$x^2 + y^2 + 6x - 4y + 4 = 0$

S3

Find the centre and radius of the circle:

$(x - 3)^2 + (y - 2)^2 = 25$

S4

Find the centre and radius of the circle:

$x^2 + y^2 - 4x - 4y - 28 = 0$

Show your working

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GEO-9D18-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Equation of a Circle

Centre and Radius

Name

E1

Find the centre and radius of the circle:

$(x + 5)^2 + (y + 1)^2 = 36$

E2

Find the centre and radius of the circle:

$x^2 + y^2 + 2x + 10y + 10 = 0$

E3

Find the centre and radius of the circle:

$(x - 6)^2 + (y - 4)^2 = 81$

KAIZEN MATHS

Exit ticket

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Centre and Radius

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Show your working

Teacher answers & working

Equation of a Circle · Centre and Radius

GEO-9D18-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Find the centre and radius of the circle:
$(x + 4)^2 + y^2 = 4$

Answer:

Centre (-4, 0), radius 2

Working:

Compare the equation with $(x - a)^2 + (y - b)^2 = r^2$.

The x bracket gives a = -4 and the y bracket gives b = 0.

The centre is (-4, 0).

$r^2 = 4$, so $r = 2$.

S2

Find the centre and radius of the circle:
$x^2 + y^2 + 6x - 4y + 4 = 0$

Answer:

Centre (-3, 2), radius 3

Working:

Group the x terms and y terms, then move the constant to the other side.

$x^2 + 6x + y^2 - 4y = -4$

Complete the square for x: $x^2 + 6x = (x + 3)^2 - 9$.

Complete the square for y: $y^2 - 4y = (y - 2)^2 - 4$.

$(x + 3)^2 + (y - 2)^2 = 9$

Centre (-3, 2), radius 3.

S3

Find the centre and radius of the circle:
$(x - 3)^2 + (y - 2)^2 = 25$

Answer:

Centre (3, 2), radius 5

Working:

Compare the equation with $(x - a)^2 + (y - b)^2 = r^2$.

The x bracket gives a = 3 and the y bracket gives b = 2.

The centre is (3, 2).

$r^2 = 25$, so $r = 5$.

Teacher answers & working

Equation of a Circle · Centre and Radius

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S4

Find the centre and radius of the circle:
$x^2 + y^2 - 4x - 4y - 28 = 0$

Answer:

Centre (2, 2), radius 6

Working:

Group the x terms and y terms, then move the constant to the other side.

$x^2 - 4x + y^2 - 4y = 28$

Complete the square for x: $x^2 - 4x = (x - 2)^2 - 4$.

Complete the square for y: $y^2 - 4y = (y - 2)^2 - 4$.

$(x - 2)^2 + (y - 2)^2 = 36$

Centre (2, 2), radius 6.

E1

Find the centre and radius of the circle:
$(x + 5)^2 + (y + 1)^2 = 36$

Answer:

Centre (-5, -1), radius 6

Working:

Compare the equation with $(x - a)^2 + (y - b)^2 = r^2$.

The x bracket gives a = -5 and the y bracket gives b = -1.

The centre is (-5, -1).

$r^2 = 36$, so $r = 6$.

E2

Find the centre and radius of the circle:
$x^2 + y^2 + 2x + 10y + 10 = 0$

Answer:

Centre (-1, -5), radius 4

Working:

Group the x terms and y terms, then move the constant to the other side.

$x^2 + 2x + y^2 + 10y = -10$

Complete the square for x: $x^2 + 2x = (x + 1)^2 - 1$.

Complete the square for y: $y^2 + 10y = (y + 5)^2 - 25$.

$(x + 1)^2 + (y + 5)^2 = 16$

Centre (-1, -5), radius 4.

Teacher answers & working

Equation of a Circle · Centre and Radius

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E3

Find the centre and radius of the circle:
$(x - 6)^2 + (y - 4)^2 = 81$

Answer:

Centre (6, 4), radius 9

Working:

Compare the equation with $(x - a)^2 + (y - b)^2 = r^2$.

The x bracket gives a = 6 and the y bracket gives b = 4.

The centre is (6, 4).

$r^2 = 81$, so $r = 9$.